INDIVIDUAL CASE SAFETY REPORT (ICSR) ASSESSMENT

Task ID: task5 | **Role:** Drug Safety Associate Simulation | **Status:** Comprehensive Evaluation Framework

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1. Patient & Case Presentation Details

Case Summary Overview: This simulated clinical scenario establishes an entry-level framework for assessing a suspected Adverse Drug Reaction (ADR). The patient, a 64-year-old female with a primary medical history of hypercholesterolemia and type 2 diabetes mellitus, was initiated on high-intensity statin therapy (Atorvastatin 40mg daily) three months prior. She presented to the clinic reporting severe, bilateral lower extremity muscle pain, weakness, and generalized fatigue lasting for the past three weeks. Laboratory investigations revealed a serum Creatine Kinase (CK) elevation exceeding five times the Upper Limit of Normal (ULN), signaling acute drug-induced myopathy.

Parameter	Patient Specification / Finding
Patient Initials / ID	PV-2026-T5
Age / Gender	64 Years / Female
Primary Indication	Hypercholesterolemia, Cardiovascular Prophylaxis
Reported Symptoms	Bilateral myalgia, proximal muscle weakness, dark tea-colored urine
Key Laboratory Findings	Serum Creatine Kinase (CK): 1,250 U/L (Normal < 150 U/L); Serum Creatinine: 1.4 mg/dL

2. Suspected & Concomitant Medication Profile

Medication Name	Daily Dose	Therapy Dates	Role in Case
Atorvastatin	40 mg PO QD	June 10, 2026 - Present	Suspected Drug (Primary)
Metformin	1000 mg PO BID	Jan 15, 2024 - Present	Concomitant (Unlikely)
Lisinopril	10 mg PO QD	Mar 22, 2025 - Present	Concomitant (Unlikely)

3. Adverse Drug Reaction (ADR) Identification

Following a comprehensive review of the clinical events, the event meets the established criteria for an authentic Adverse Drug Reaction. The presentation displays a strong temporal pattern, arising 10 weeks following a dose escalation of the primary lipid-lowering medication. Differential diagnoses including primary polymyositis, hypokalemia, and viral-induced myositis were systematically ruled out via negative antibody screens and unremarkable viral panels.

4. Standardized Causality Assessment (Naranjo Algorithm)

To establish a clear objective assessment of probability, a standardized Naranjo Scale evaluation was conducted by the Drug Safety Associate. The detailed scoring matrix is summarized below:

Naranjo Assessment Questionnaire Criteria	Response	Score
1. Are there previous conclusive reports on this reaction?	Yes	+1
2. Did the adverse event appear after the suspected drug was administered?	Yes	+2
3. Did the adverse reaction improve when the drug was discontinued?	Yes (Dechallenge)	+1
4. Did the adverse event reappear when the drug was readministered?	Not Rechallenged	0
5. Are there alternative causes that could on their own have caused the reaction?	No	+2
6. Did the reaction reappear when a placebo was given?	No	+1
7. Was the drug detected in blood or other fluids in toxic concentrations?	Not Measured	0
8. Was the reaction more severe when the dose was increased?	Yes	+1
9. Did the patient have a similar reaction to the same or similar drugs?	No History	0
10. Was the adverse event confirmed by any objective evidence?	Yes (CK Level)	+1
TOTAL NARANJO SCORE	**8 / 10**	**PROBABLE**

5. Severity & Seriousness Case Classification

Under current **ICH E2A guidelines**, a clear distinction must be made between clinical severity and regulatory seriousness:

• **Clinical Severity Grade:** Severe (Grade 3 according to CTCAE criteria due to profound impact on standard activities of daily living and severe laboratory abnormalities).

• **Regulatory Seriousness:** Classified as **Serious** due to the onset of medically significant clinical risks. The presence of dark tea-colored urine alongside elevated serum creatinine alerts to an impending risk of acute rhabdomyolysis-induced renal impairment, requiring immediate therapeutic intervention and drug discontinuation.

• **Expedited Reporting Requirement:** As a Serious, Expected ADR, this case is tracked under routine regulatory timelines (15-day alert report) rather than an expedited 7-day SUSAR timeline.

6. Drug Safety Associate Conclusion & Action Plan

Conclusion: A definitive causal link is established connecting Atorvastatin therapy to severe drug-induced myopathy (Naranjo Score 8: Probable).

Immediate Clinical Action Recommendations:

1. Permanent discontinuation of Atorvastatin therapy immediately.
2. Initiate aggressive intravenous fluid hydration to safeguard renal clearance and monitor urinary output to mitigate rhabdomyolysis risk.
3. Re-evaluate serum Creatine Kinase (CK) and serum creatinine every 48 hours until values return to patient baseline.
4. Consider alternative non-statin lipid-lowering modalities (e.g., Ezetimibe or PCSK9 inhibitors) once clinical symptoms resolve completely.

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